Fiscal Year:	FY 2012	**Task Last Updated:**	FY 06/05/2012
PI Name:	Stone, Lee S. Ph.D.		
Project Title:	Robust Human-System Interface Design for Spaceflight Induced Environments		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture (2) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	650-604-3240
Organization Name:	NASA Ames Research Center		
PI Address 1:	Human-Systems Integration Division		
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City:	Moffet Field	**State:**	CA
Zip Code:	94035-1000	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Crew Health NNJ08ZSA002N
Start Date:	05/01/2009	**End Date:**	04/30/2013
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kaiser, Mary (NASA Ames Research Center) Cowings, Patricia (NASA Ames Research Center) Toscano, Bill (NASA Ames Research Center) Sweet, Barbara (NASA Ames Research Center) Adelstein, Bernard (NASA Ames Research Center) Liston, Dorion (NASA Ames Research Center)		
Grant/Contract No.:	NCC 9-58-SA02002		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The next generation of human-rated spaceflight vehicles will expose astronauts to altered gravito-inertial forces and elevated vibration levels during launch and landing that can affect their ability to perform their assigned tasks. Furthermore, current plans call for crew action in critical tasks, including the override of flight automation in off-nominal conditions, while the crew may be exposed to 4-times normal gravity or more combined with vibration levels potentially higher than those experienced during Apollo and Shuttle. The risk associated with these dynamic phases of flight becomes even more acute when critical tasks are performed by de-conditioned crew returning after a long-duration spaceflight. Altered gravitational environments experienced in spacecraft impact human manual and gaze control and high-frequency vibration is also a serious challenge to visuomotor performance and gaze control. Unfortunately, our current quantitative knowledge of these effects is too limited to develop and validate operationally relevant predictive models of crew performance of sufficient reliability to support interface design or requirement verification. In particular, the interactions between increased gravity and vibration are poorly understood. The goal of the proposed research is to measure, analyze, and model human performance in display navigation and flight control tasks during centrifugation and vibration (in isolation and combination) to simulate ascent/descent conditions of spaceflight. The proposed research will measure performance (and associated workload) in operationally relevant display navigation and manual control tasks under simulated dynamic phases of flight and will compare the effectiveness of a variety of potential display and interface configurations. These data will enable us to identify and model the human capabilities and limitations in order to provide validated quantitative design guidelines for displays and controls to support crew-cockpit design engineers and operations/training development. This research will enable the development of validated spacecraft interface design guidelines and human-system verification tools with the ultimate goal of reducing mission risks and enhancing safety.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed research will determine the impacts of transient and sustained acceleration on human sensorimotor performance. The benefit of this research for NASA is that it will fill a critical knowledge gap that is currently hampering the optimal design of astronaut and pilot displays and interfaces needed to maximize performance and safety and minimize human error. The benefit of this research for humanity above and beyond enabling safer and more productive space exploration is two-fold. First, the human performance databases (and models) generated will allow for better design of aircraft and other vehicles thereby ultimately facilitating the enhancement of safety for the travelling public by supporting improved cockpit/interface design and training. Second, the performance assessment technologies developed and validated in this project may have medical applications for enhancing diagnostic methods to identify sensorimotor deficits due to disease or injury.
Task Progress:	This year, we completed two studies (12 human subjects in each) measuring human performance under +1 and +3.8 Gx sustained loads into the chest with an added ±0.5g vibration at three different vibration frequencies (10, 12, and 18 Hz) in paired experiments within our human vibration laboratory (+1 Gx experiments) and the Ames 20g centrifuge (+3.8 Gx experiments), each equipped with matched, wide-field, multi-display touchscreen units and embedded high-speed high-resolution video head/eye tracker. In the first study, we measured gaze control and rapid reach performance in a single task allowing the measurement of vestibular function (gaze stabilization during the initial fixation), visual spatial localization (gaze acquisition to a flashed peripheral target), and manual control (rapid reaches to the same target). Our initial analyses of the reaching data show that: - 3.8-Gx loading increases the systematic error of the head-to-foot component of manual reaches up to five-fold (largest errors for lower ipsilateral targets) but surprisingly improves the accuracy of the lateral component, compared to performance at 1 Gx; - 3.8-Gx loading decreases the precision of both directional components of manual reaches (increases error dispersion by ~20%), compared to that at 1 Gx; and - Added vibration decreases the precision of lateral component of reach regardless of Gx bias (increases error dispersion by ~10%) with little effect of vibration frequency across the tested range. Analysis of the eye-movement data is ongoing, but preliminary analyses have found a significant correlation (Pearson's r, $p < 0.01$) between the lateral errors in eye and reach end points with at least 20% of the variance in error due to shared neural noise in visual localization. In the second study, we measured continuous gaze control in a target-tracking task allowing the measurement of visual motion processing (smooth pursuit of a moving target spot). Data analysis of this experiment has just begun. Our studies are providing the first quantitative measures of human vestibular, visual, and manual control performance under a range of sustained Gx and vibration loads similar to those contemplated during the launch of SLS and commercial space vehicles. These trade-space data will help guide the design of operations concepts and interfaces used during the dynamic phases of flight and the development of more complete models of human performance in exploration-relevant conditions.
Bibliography Type:	Description: (Last Updated: 10/30/2019)
Abstracts for Journals and Proceedings	Liston DB, Adelstein BD, Beutter BR, Stone LS. "New method for quantifying human high-frequency linear and rotational VOR during launch-relevant conditions." Eighth Symposium on the Role of the Vestibular Organs in Space Exploration, Houston, TX, April 8–10, 2011. Journal of Vestibular Research, 2011;21(2):84. , Apr-2011